PROGRAM 5 – MISSIONARIES AND CANNIBALS PROBLEM

AIM

To write a Python program to solve the Missionaries and Cannibals problem using Breadth First Search.

ALGORITHM

1. Start with three missionaries and three cannibals on the left bank.
2. Keep the boat on the left bank initially.
3. Allow the boat to carry one or two people.
4. Generate all possible boat movements.
5. Check that cannibals never outnumber missionaries on either bank when missionaries are present.
6. Store valid states in a queue.
7. Continue until all missionaries and cannibals reach the right bank.
8. Display the solution path.

CODE

```
from collections import deque
def safe(m,c):
    if m<0 or c<0 or m>3 or c>3:
        return False
    if m>0 and m<c:
        return False
    mr=3-m
    cr=3-c
    if mr>0 and mr<cr:
        return False
    return True
def solve():
    start=(3,3,0)
    goal=(0,0,1)
    moves=[(1,0),(2,0),(0,1),(0,2),(1,1)]
    queue=deque([(start,[start])])
    visited={start}
    while queue:
```

```
        state,path=queue.popleft()
        if state==goal:
            return path
        m,c,boat=state
        for dm,dc in moves:
            if boat==0:
                new_state=(m-dm,c-dc,1)
            else:
                new_state=(m+dm,c+dc,0)
            if safe(new_state[0],new_state[1]) and new_state not in visited:
                visited.add(new_state)
                queue.append((new_state,path+[new_state]))
    return None
solution=solve()
print("Solution:")
for state in solution:
    print(state)
```

INPUT

```
start=(3,3,0)
goal=(0,0,1)
moves=[(1,0),(2,0),(0,1),(0,2),(1,1)]
```

OUTPUT

```
Solution:
(3, 3, 0)
(3, 1, 1)
(3, 2, 0)
(3, 0, 1)
(3, 1, 0)
(1, 1, 1)
(2, 2, 0)
(0, 2, 1)
(0, 3, 0)
(0, 1, 1)
(1, 1, 0)
(0, 0, 1)
```

RESULT

Thus, the Missionaries and Cannibals problem was successfully solved using Breadth First Search.